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To cite this article: H Anand *et al* 2021 *Plasma Phys. Control. Fusion* **63** 015006

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Plasma flux expansion control on the DIII-D tokamak

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Received 9 September 2020, revised 8 October 2020

Accepted for publication 23 October 2020

Published 13 November 2020



Abstract

A new controller has been developed with help of the flexible divertor poloidal-field coil set of the DIII-D tokamak, to aid in the precise control of the flux expansion in the scrape-off layer. The single-input multiple-output architecture ensures flexibility through a complementary set of orthogonal actuator direction to guarantee minimum effect on existing controlled variables, e.g. radial and vertical position control of the X-point. A non-linear free-boundary simulation code (GSevolve) is used for simulating the closed-loop response and for verifying the implementation of the control algorithm on the DIII-D plasma control system. First results of the experimental commissioning of the new controller during 2020 DIII-D campaign are also presented.

Keywords: real-time control, magnetic control, scrape-off layer, flux expansion, magnetic equilibrium reconstruction.

(Some figures may appear in colour only in the online journal)

1. Introduction

The capability of plasma-facing components (PFCs) to withstand the flux of energy and particles from the confined plasma is one of the most severe challenges for the design of future magnetic fusion reactors. Fusion power plants based on the tokamak concept will therefore likely require the expulsion of a high fraction of the plasma heating power as radiation and the establishment of a pressure gradient along the SOL field lines, characteristic of operation in a detached divertor regime aspects [1]. The research presented in the paper aims at improving one of the key aspect influencing this regime, i.e. the magnetic configuration [2]. The magnetic geometry in the divertor can be modified by changing the poloidal flux expansion at the target, $f_{x,t} = dr_t/dr_{omp}$, where dr_t and dr_{omp} are the distance between the same flux surfaces at the divertor target and at the outer-midplane (OMP), respectively (figure 2). Increasing $f_{x,t}$, increases the wetted area and, hence, decreases the peak heat load on the targets [2–4]. However, an increased flux expansion reduces the angle between the magnetic field lines and the target plate, which cannot be arbitrarily small [5]. Due to mechanical alignment tolerances of the divertor target plates, the field line angle is limited to values above

approximately 1° . In addition to a geometric reduction of the heat flux onto the target, a greater wetted area is predicted to increase the interaction with neutrals, larger connection length at the target, $L_{||,t}$, and greater SOL volume, V_{SOL} [6]. Recent research [2] indicates the degree of detachment, or the location of the detachment front [7] is dependent on flux expansion for advanced plasma configurations. In existing devices and for future ones, detachment control primarily relies on the slow response (1–2 s) of the gas valves for feedback control and can cause severe damage to the PFCs upon exceeding their critical heat flux limits. Although flux expansion is not expected to be a very strong actuator (it should not be relied upon to achieve and hold detachment by itself), it should be very fast: changes in flux expansion are expected to be effective on the timescale of the ion equilibration time in the SOL ($\approx$ few ms), significantly faster than gas control (1–2 s). Flux expansion control offers the possibility of supplementing gas as an actuator for detachment control and improving the response time of the controller. That is, a disturbance in the plasma would warrant a change in gas command which will take effect some time later. While waiting for the plasma to respond to the gas command, the flux expansion control can temporarily counteract a minor or moderate disturbance

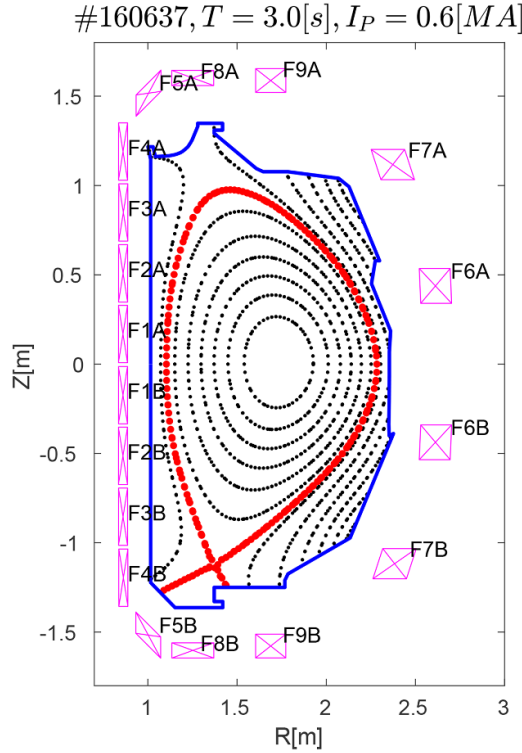


Figure 1. Poloidal cross-section of the DIII-D tokamak showing the flux surfaces obtained from RTEFIT.

to protect the divertor target plate. Thus, robust, reliable and simplified RT models for monitoring and effective feedback control strategies for plasma flux expansion in the SOL is of utmost importance in plasma boundary physics [2–4]. So far, EAST has been the only tokamak that has successfully demonstrated the real-time flux expansion control based on a multi-input multi-output (MIMO) architecture in order to achieve integrated control of both plasma boundary and flux expansion in the SOL [8]. In comparison with the previous published work, the novel flux expansion controller developed for the DIII-D tokamak is based on a SIMO architecture that allows for easier implementation and overcomes the interactions between the actuators through the identification of orthogonal directions. These features help to provide a user-friendly interface that can be used routinely during plasma operations for reliable control of the flux expansion for divertor plasma configuration.

Until now on DIII-D, the plasma flux expansion has been controlled almost entirely in feedforward mode. Deviations from the assumptions used in calculating the feedforward parameters can and do cause departures from the desired flux expansion in the absence of feedback control. This happens in particular as a result of auxiliary heating and current drive altering the current profile. Feedback control of flux expansion is also clearly superior to pure feedforward control in its ability to deal with disturbances. Success in controlling flux expansion in feedforward mode in DIII-D has often been achieved by virtue of tuning over multiple discharges.

The control of plasma flux expansion control in the SOL involves two steps:

- Estimation of the plasma flux expansion in real-time.
- Adjustment of the poloidal field (PF) coil currents to bring the real flux expansion as close as possible to the pre-programmed value.

The presence of the real-time version of the Grad–Shafranov equilibrium reconstruction code EFIT, i.e. RTEFIT [9], with a sub-millisecond calculation time, has fulfilled a necessary requirement for the development of a real time plasma flux expansion estimation algorithm, based on the information of poloidal flux and magnetic field provided by the real-time Grad–Shafranov solver. RTEFIT and other similar real time equilibrium codes [10] provides an approximate solution to the Grad–Shafranov equilibrium relation that yields the best fit to the diagnostic measurements, with additional assumptions on the plasma kinetic profiles, as the diagnostic data are generally insufficient to constrain the problem toward a unique solution.

The main operational parameters of the DIII-D tokamak are the following: typical major radius 1.69 m, typical minor radius 0.60 m, vacuum toroidal field up to 2.2 T, plasma current up to 2 MA, elongation up to 2.0 [11]. Figure 1 shows the DIII-D poloidal cross-section with lines of constant flux shown to illustrate the plasma magnetic equilibrium. The primary purpose of the Ohmic heating coil (E-coil) solenoid is to induce current in the plasma through transformer action (figure 1 in [11]). This heats the plasma through resistive losses and provides a distributed current which can be shaped by external currents. The 18 PF coils (F-coils) F1A to F9B are used to control the shape and position of the plasma. Shaping power supplies act as voltage sources in series with certain of the F-coils to provide the necessary shaping currents.

This extensive PF coil system and the presence of RTEFIT on DIII-D have paved the way to the design of a new flux expansion controller. In addition, the Embedded Matlab Coder (EMC) environment of the DIII-D plasma control system (PCS) [12, 13] provides a flexible platform for implementing new algorithms programmed with Matlab/Simulink.

The paper is structured as follows. Section 2 describes the flux expansion estimator. Section 3 describes the design and the key features of the new flux expansion controller. The closed-loop performance of the controller and its verification on the DIII-D PCS, using GSevolve [14] is also reported in section 3. First experimental results on the flux expansion controller during plasma operations are presented in section 4. A brief overview of the controller design, modeling and experimental results, as well as an outlook for the physics applications of the controller is provided in section 5.

2. Estimation of flux expansion in the SOL

The flux expansion, $f_{x,t}$, is the separation distance dr between two flux surfaces in the SOL at a location of interest (here, the divertor target: dr_t) normalized to their separation at a reference location, such as the OMP dr_{omp} . Figure 2 shows how the distance dr between a flux surface and the separatrix varies along the SOL.

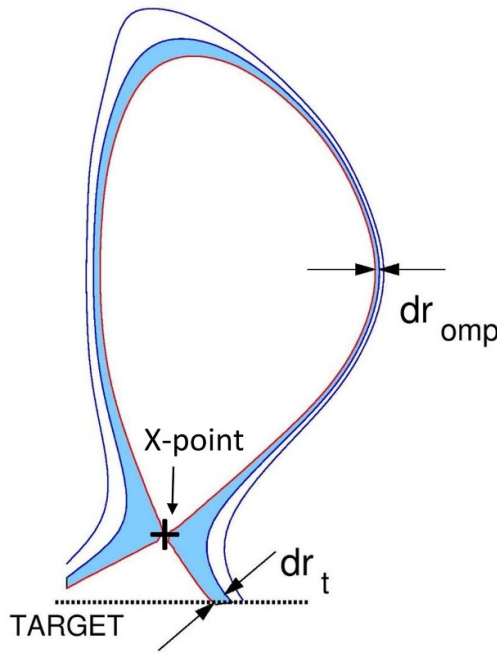


Figure 2. Schematic showing the distance dr between two flux surfaces along the SOL.

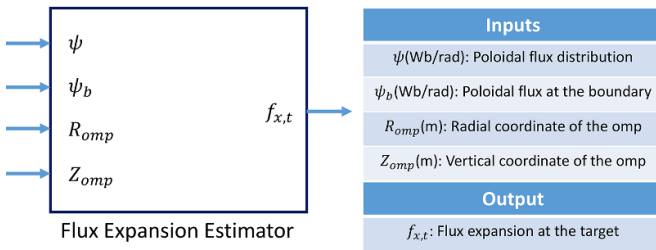


Figure 3. Matlab/Simulink schematic for flux expansion estimation and definition of input/output data for the application on DIII-D.

The difference in poloidal magnetic flux between adjacent flux surfaces, $d\psi$, can be expressed as $d\psi = |\nabla\psi| dr$, $|\nabla\psi| = RB_\theta$, where B_θ is the poloidal magnetic field, yields $d\psi = RB_\theta dr$. The flux expansion is defined as the change in the distance dr between two flux surfaces relative to that at a reference location in the SOL. By convention, the distance dr at the upstream location is often used as the reference since radial fall-off lengths at the upstream location are largely determined by the core plasma parameters. For example, the target flux expansion is given by:

$$f_{x,t} = \frac{dr_t}{dr_{omp}} = \frac{R_{omp}B_{\theta,omp}}{R_tB_{\theta,t}}, \quad (1)$$

where $B_{\theta,omp}$ and $B_{\theta,t}$ are the magnitudes of the poloidal magnetic field at the OMP (R_{omp}, Z_{omp}) and target (R_t, Z_t) locations, respectively.

The Matlab/Simulink implementation of the algorithm, accommodating various inputs and outputs is shown in figure 3, along with a description of the inputs/outputs used for the application to DIII-D. The Simulink model is implemented on a dedicated CPU of the DIII-D PCS with the help of the

EMC feature and cycles at a sample time of 3 ms. The RTEFIT provides the information at sub-ms time steps, 0.25 ms for the poloidal flux distribution, ψ , poloidal flux at the last closed flux surface or separatrix, ψ_b , R_{omp} and vertical coordinate of the OMP, Z_{omp} . It must be noted that the computational time for the flux expansion estimation algorithm is negligible (≈ 0.1 ms) and is largely governed by the computational time of the RTEFIT (≈ 1 – 2 ms). Then, the resulting commands produce a response in the PF coil currents on a multi-ms timescale (≈ 10 – 15 ms). Thus, the small additional time delay due to the flux expansion estimation does not influence the overall magnetic control loop.

3. Implementation and controller design

Figure 4 shows an overview block diagram of the isoflux plasma shape control [9] presently used in DIII-D. Plasma diagnostics acquired by the PCS in real time are used to make an estimate of the plasma equilibrium shape and current distribution, from which the X-point location and control point flux errors are calculated. These errors are processed to produce commands to shape control power supplies on the plasma shaping coils. A fast, vertical stability control algorithm is also executed within the PCS. This controller does not fully stabilize the plasma, since it has no proportional feedback term; instead it reduces the growth rate sufficiently that the slower shape control algorithm can stabilize the plasma.

The controlled variables are the values of flux at pre-specified control points on the plasma boundary, along with the X-point radial position, R_X and vertical position, Z_X . This ‘isoflux’ scheme applies PID gains to control point flux and X-point position errors. The resulting PID signals are multiplied by a gain matrix, M to produce commands to the power supplies on each plasma shaping coil. The gain matrix is sparse, such that, most individual shape errors are corrected through the application of only a small number (often one) of coil current changes. The control of the X-point position requires coordinated action by a subset of divertor shaping coils. A detailed description of the DIII-D PCS architecture can be found in [9].

The implementation of the new flux expansion controller with the DIII-D isoflux controller is also shown in figure 4. In addition to the M matrix, the column vector for flux expansion, $\vec{M}_f$ selects the combination of PF coils orthogonal to the coil combinations providing vertical, $\vec{M}_{Z_X}$ and radial control, $\vec{M}_{R_X}$ of the X-point. More details on the construction of the M matrix can be found in [15]. The reference signals for the flux expansion, $f_{x,t,ref}$ are user-defined. The error signals, $e_f = f_{x,t,ref} - f_{x,t,mes}$ are fed to a proportional and integral controller with a continuous-time transfer function, $K_f(s)$, such that,

$$\vec{u}_f = K_f(s)e_f, K_f(s) = K_p \left(1 + \frac{K_i}{s} \right) \quad (2)$$

where, $\vec{u}_f$ is the output of the controller and K_p and K_i are the proportional and integral gain that are to be designed.

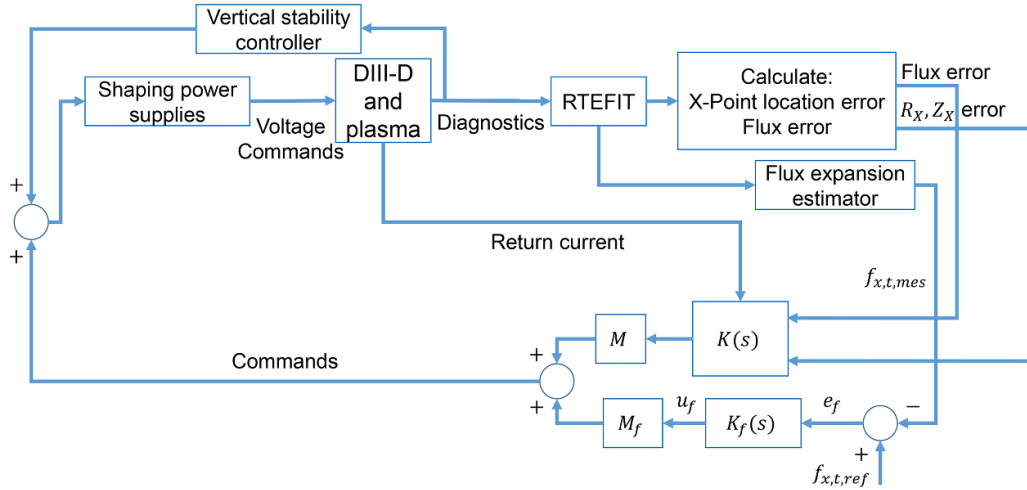


Figure 4. Block diagram showing the implementation of the flux expansion controller with the DIII-D isoflux controller.

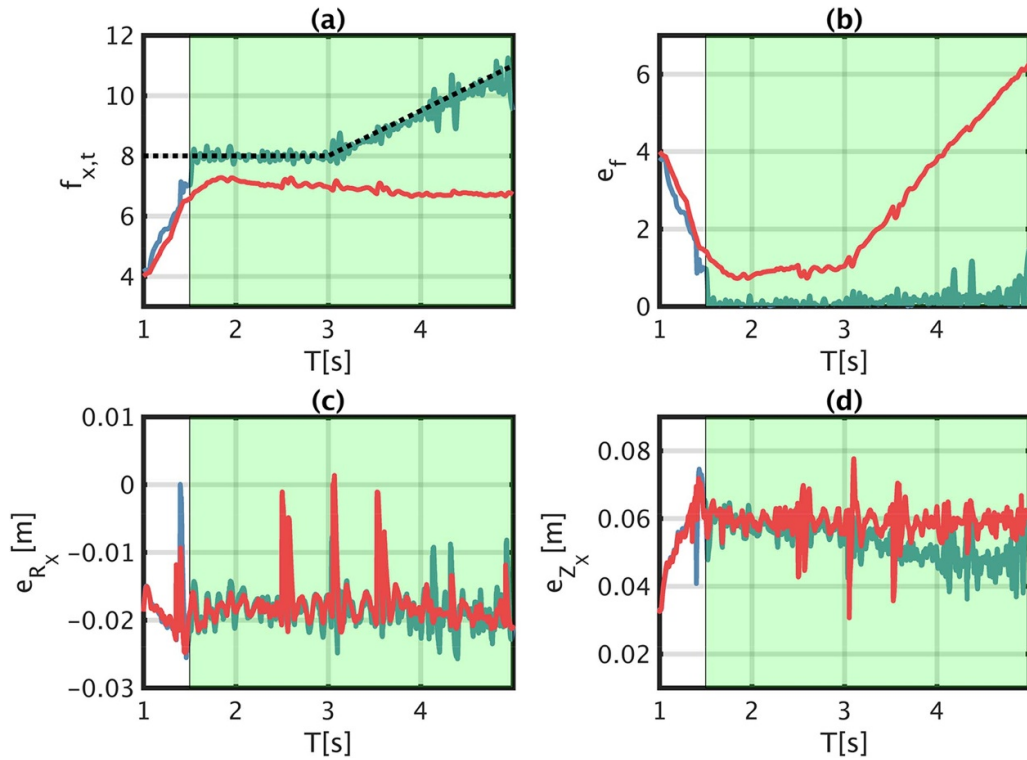


Figure 5. Simulation comparing the performance of the DIII-D isoflux controller (sim #967237, solid red) and the optimized flux expansion controller (sim #967236, solid blue). (a) Time evolution of the $f_{x,t}$ and the pre-programmed reference (dotted black). (b)–(d) Time evolution of the error in $f_{x,t}$, R_X and Z_X . The activation time interval of the flux expansion controller is shown by the shaded region in green.

The tuning of the control parameters and the verification of the flux expansion controller is performed with the help of a non-linear free-boundary simulation code, GSevolve [14]. This simulation code evolving the Grad–Shafranov equilibrium including current and pressure profiles resides within the suite of tools, TokSys [16], developed at DIII-D over the past decade to support integrated control design. The unique capability of the software suite to directly connect to the DIII-D PCS, also termed as ‘simulation server’ (simserver) [12, 13] provides simulated tokamak and plasma data in response to

the control commands from the flux expansion controller. The closed-loop simserver simulations help to identify the control parameters and confirm the implementation and operation under realistic hardware/software conditions.

GSevolve simulations are shown in figure 5 for the reference plasma configuration defined in figure 1. The controller performance is compared with the results achieved using the DIII-D isoflux controller, which merely provides feedforward control for the flux expansion. For comparing the controller performance, simulation with optimized controller setup

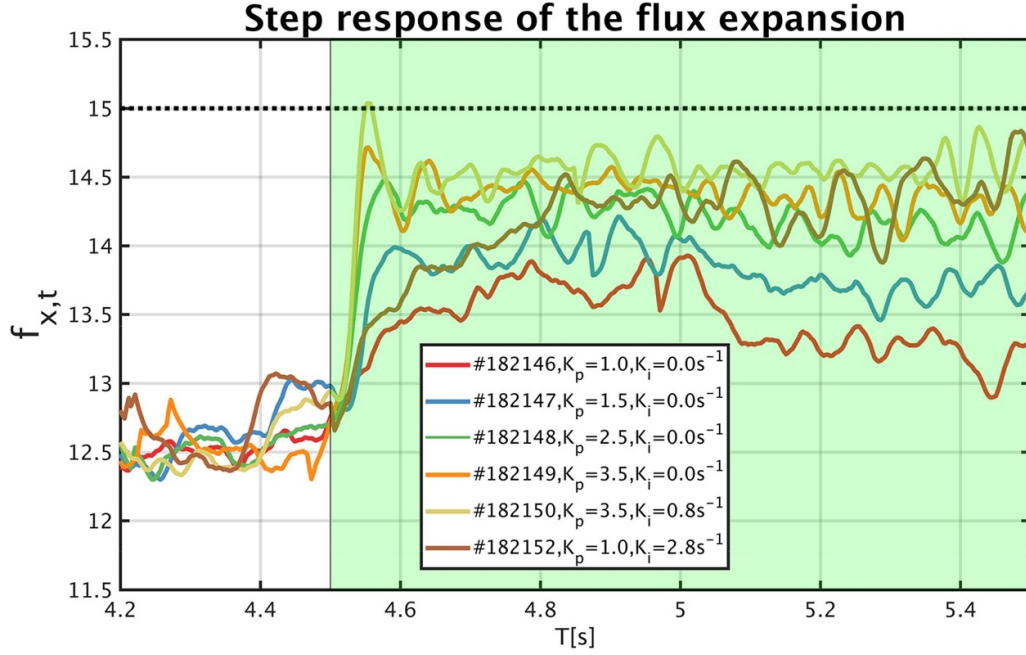


Figure 6. Step response of the flux expansion for different plasma discharges with their associated values of K_p and K_i . The pre-programmed reference for the flux expansion is shown by the dotted black line and the activation time interval of the flux expansion controller is shown by the shaded region in green.

(sim #967236), with a choice of $K_p = 3.5$ and $K_i = 0.8 \text{ s}^{-1}$, based on the reference discharge 160 637 (which did not have the new optimized controller setup) is performed. The parameters K_p and K_i were optimized through scans of their values in GSevolve simulation, simply to provide initial values as a starting point for the experiment. The parameters were then optimized further through experimental scans of their values, as shown in figure 6.

The same reference was then simulated with the DIII-D isoflux controller (sim #967237). The performance of the controller is evaluated by studying the response of the flux expansion for an arbitrary pre-programmed reference waveform and its effect on the control of X-point position. The PF coils F5B, F8B and F9B were used to control the flux expansion and the respective values of the column vectors constituting a subset of the M matrix, i.e. the $\vec{M}_f$, $\vec{M}_{R_x}$ and $\vec{M}_{Z_x}$ for flux expansion control in the SOL and position control of the X-points are given as follow:

$$\vec{M}_f = \begin{bmatrix} +330 \\ -907 \\ +264 \end{bmatrix}, \quad \vec{M}_{R_x} = \begin{bmatrix} -200 \\ 0 \\ +250 \end{bmatrix}, \quad \vec{M}_{Z_x} = \begin{bmatrix} -75 \\ -100 \\ -250 \end{bmatrix}. \quad (3)$$

Figure 5(a) shows the pre-programmed ramp reference defined by the user and the measurement of the $f_{x,t}$ for the two simulated discharges. The new controller provides better tracking of the $f_{x,t}$ and also leads to low steady state error in comparison to the plasma discharge controlled by the DIII-D isoflux controller. As shown in figure 5(b), during the constant and ramp-up phase of the $f_{x,t}$ operation, $\approx$ zero steady state error value was obtained for the sim #967236 associated with the new controller in comparison to a finite value of for sim

#967237 with the DIII-D isoflux controller. Figures 5(c) and (d) illustrate the difference between the errors in R_x and Z_x for the two discharges. For the sim #967236 performed with the flux expansion controller, the error in R_x and Z_x increases and diverge slightly from the values obtained during the plasma discharge, whereas the errors remained almost constant for sim #967237 with a finite steady state error. The root mean square deviation between the errors in R_x and Z_x for the two discharges shown in figures 5(c) and (d) are 0.004 m and 0.008 m, respectively. Thus, for the $\vec{M}_f$ used for flux expansion control, a suitable range of flux expansion (8–11) for a fixed X-point location was successfully controlled with a minimal effect on the X-point position.

4. Experimental results

Dedicated experiments have been performed to test the new flux expansion controller during the DIII-D campaign in 2020.

Divertor plasma discharges (#182146–182150 and #182152) involving fixed position of the outer strike point (OSP) in time were chosen to test the controller formalism with the aim of exploring the range of K_p and K_i for controlling the flux expansion in the SOL with the help of the PF coils, F5B, F8B and F9B. The fixed set of column vectors, $\vec{M}_f$, $\vec{M}_{R_x}$ and $\vec{M}_{Z_x}$ used in the discharges are given by equation (4). Figure 6 illustrates the range of control parameters that provide suitable performance in conjunction with the position controller of the X-point of the isoflux DIII-D controller. The median range of consistent proportional (1–3.5) and integral gains (0–2.8 s^{-1}) for a given divertor plasma configuration is shown in figure 6.

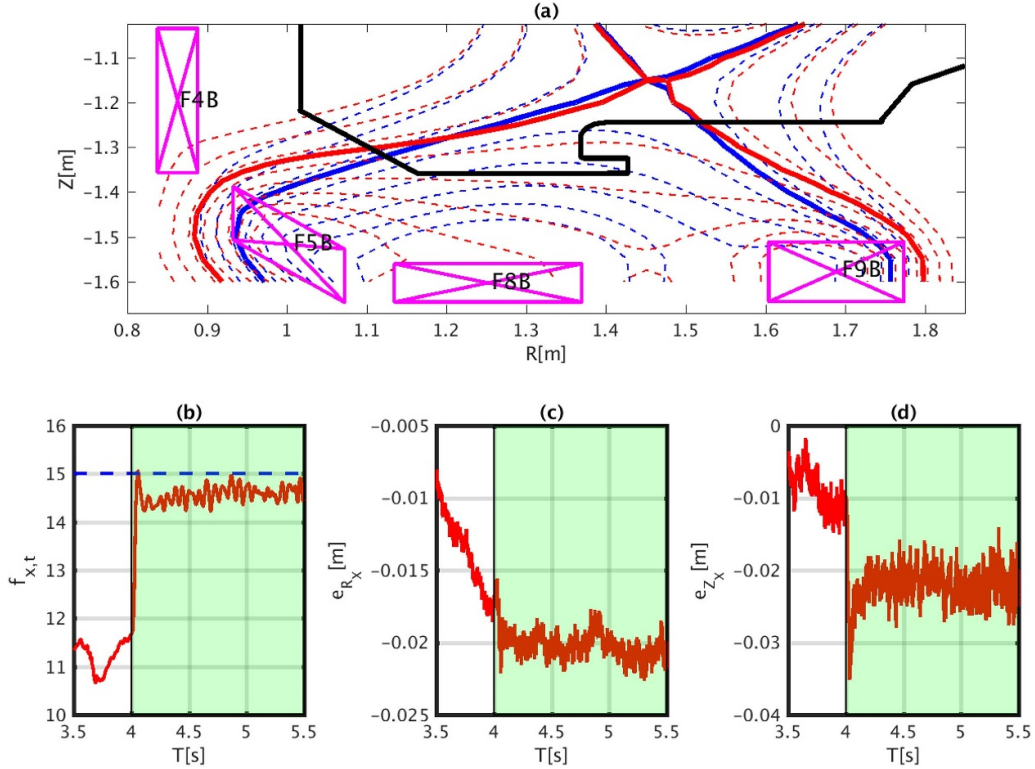


Figure 7. Performance of the flux expansion controller for divertor plasma configuration involving fixed position of the OSP. (a) Difference in the poloidal flux contour (dotted line) and the separatrix (solid line) before ($t = 3.9$ s, blue) and after the controller operation ($t = 5.5$ s, red) obtained from RTEFIT. (b)–(d) Time evolution of the error in $f_{x,t}$, R_X and Z_X . The activation time interval of the flux expansion controller is shown by the region shaded in green.

For demonstrating the feature of the controller design of providing a complementary set of orthogonal actuator direction to guarantee minimum effect on radial and vertical position control of the X-points, a dedicated divertor plasma discharge with fixed position of the OSP was perturbed (#182153) with a given set of fixed controller gains ($K_p = 3.5$, $K_i = 0.8 \text{ s}^{-1}$) and values of column vectors in the M matrix given by equation (4).

$$\vec{M}_f = \begin{bmatrix} +503 \\ -703 \\ +503 \end{bmatrix}, \vec{M}_{R_X} = \begin{bmatrix} -250 \\ 0 \\ +200 \end{bmatrix}, \vec{M}_{Z_X} = \begin{bmatrix} -250 \\ -250 \\ -100 \end{bmatrix}. \quad (4)$$

Figure 7(a) show the contour plots of the poloidal flux distribution and the separatrix, at time instants before and after the controller activation for the discharge #182153. They illustrate the enhancement in the plasma flux expansion in comparison to the DIII-D isoflux controller for a given set of optimized control parameters. As per the definition of the plasma flux expansion given in figure 2, figure 7(a) show a larger distance between the separatrix (solid line: red) and the adjoining poloidal flux surfaces (dotted line: red) and hence the presence of a larger plasma flux expansion at the divertor target after the controller activation in comparison to the distance between the separatrix (solid line: blue) and the adjoining poloidal flux surfaces (dotted line: blue) before the onset of the controller. Figure 7(b) illustrates a good tracking of the pre-programmed reference of the plasma flux expansion in the

SOL with the activation of the controller. Figures 7(c)–(d) explores the effect of the flux expansion controller on the error of the controlled variables representing the radial and vertical position of the X-points. During the controller operation, the error in R_X is held constant with a finite steady state value ≈ 2 cm, whereas a finite change ≈ 1 cm is observed in the error in Z_X before and after the controller activation. In particular, the impact of the new controller on the existing control of the X-point position is minimal and the small variations, especially in Z_X does not impact physics studies on DIII-D.

5. Summary and outlook

A real-time flux expansion control algorithm, developed particularly for future physics application on DIII-D tokamak, has been presented in this paper. The paper discusses the modeling, design and first experimental results of a new flux expansion controller for divertor plasma configuration. A non-linear free boundary simulation code (GSevolve) is used to study the coupling between the DIII-D isoflux controller and the new flux expansion controller. The model is also used for the verification of the control algorithm and identification of the control parameters on the DIII-D PCS. A successful experimental implementation of the control algorithm has been demonstrated for divertor plasma discharges. The control design exhibits an improved performance with respect to the

control of the plasma flux expansion relative to the isoflux DIII-D controller. A suitable range of controller gains has been found through simulation as well as during experiments which provides good control of the plasma position without exciting disturbances, particularly in the position control of the X-points. The new controller is able to successfully achieve and track flux expansion values that covers the typical range of values necessary for physics studies on DIII-D tokamak. In particular, the impact of the new controller on the existing control of radial and vertical position of the X-point is minimal, thanks to the complimentary set of orthogonal actuator directions.

In spite of considerable success in SOL physics experiments without active control of plasma flux expansion, more advanced future developments in DIII-D are likely to depend on reliable flux expansion control. In the upcoming DIII-D campaign 2020–2021, dedicated experiments with the controller are planned to study the effect of the plasma flux expansion on the density required for the onset of detachment and on the evolution of detachment radiation front from the target toward the X-point.

The controller formalism is also planned to be extended on MAST-U, an upgrade of the original MAST tokamak that includes large and well-diagnosed divertor chambers with additional PF shaping coils [17]. A large part of the research plan on MAST-U focuses on exploiting the divertor chambers, including the wide range of flux expansion made possible by the upgrade [18]. The simplified design of orthogonal actuators provided by the new controller is being considered for providing reliable and independent control of flux expansion while having minimal effect on the other magnetic controlled variables in the MAST-U divertor geometry.

5. Acknowledgment and Disclaimer

Acknowledgment: This material is based upon work supported by the U S Department of Energy, Office of Science, Office of Fusion Energy Sciences, using the DIII-D National Fusion Facility, a DOE Office of Science user facility, under Award(s) DE-FC02-04ER54698.

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